Omar Abdelhamid

AI Engineer | Agentic Systems, Advanced RAG, Reasoning & Distributed MLOps

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PROFESSIONAL SUMMARY

Software-driven AI Engineer with a strong track record across highly selective industry R&D internships, prestigious tech fellowships (DEBI, NTI), and 4x AWS Certifications. Hands-on experience architecting production-grade multi-agent systems, optimizing hybrid retrieval (Dense + GraphRAG) pipelines, and deploying high-concurrency model inference (vLLM, Triton) on Kubernetes and AWS. Currently pursuing an M.Sc. in Artificial Intelligence while actively conducting applied SLM reasoning and post-training alignment research at Orange Innovation Labs. Dedicated to bridging academic research with reliable enterprise infrastructure.

WORK & INTERNSHIP EXPERIENCE

Orange Innovation Egypt **Jul 2026 – Present**
AI R&D Intern *Cairo, Egypt*
- Conducting applied AI research on Code SLM reasoning vs. shortcut memorization under Dr. Ghada Soliman; formulated the *Reduction Ladder* diagnostic and designed novel *Inv-GRPO* mitigation [Codebase].
- Evaluating open-weight LLMs, RLVR alignment, and multi-agent reasoning workflows for enterprise telecommunications, optimizing for token efficiency, latency bounds, and deterministic execution on edge infrastructure.

National Telecommunication Institute (NTI) **Jan 2026 – April 2026**
Agentic AI Engineer Trainee *6th of October, Egypt*
- Engineered a production-ready LangGraph multi-agent pipeline leveraging ReAct reasoning loops, integrated with ColPali for robust multimodal document retrieval.
- Architected a fault-tolerant 4-layer microservices backend featuring human-in-the-loop (HITL) checkpoints via AsyncRedisSaver and self-healing SQL query generation.
- Established comprehensive system observability utilizing Prometheus and Grafana to track p50/p95/p99 inference latency, ensuring scalable performance during peak loads.

Orange Digital Center Egypt – Level 3 **Nov 2024 – Jan 2025**
AI Engineer Intern (Advanced Track) *Cairo, Egypt*
- Deployed a real-time YOLOv8 pose estimation pipeline on edge infrastructure, maintaining a stable 30 FPS inference rate for seamless user interaction.
- Fine-tuned BERT embeddings for domain-specific text classification, achieving 92% accuracy against live data to automate categorization and reduce manual processing time.
- Developed a GAN-based data augmentation pipeline (DCGAN, VAE, WGAN), improving visual synthesis quality by 40% (measured via FID) to balance scarce datasets effectively.

Digital Egypt Builders Initiative (DEBI) **Jun 2024 – Oct 2024**
AWS Machine Learning Fellow (MCIT Scholarship) *Cairo, Egypt*
- Provisioned and orchestrated scalable MLOps workflows on AWS SageMaker, integrating MLflow and Hugging Face to standardize model registries across teams.
- Automated the full ML lifecycle—from distributed training (DDP) to highly available REST API deployment—significantly reducing time-to-market for AI features.

IT Gate Academy **Dec 2023 – Sep 2024**
DevOps & Cloud Infrastructure Trainee *Cairo, Egypt*
- Maintained automated CI/CD pipelines (Jenkins, GitHub Actions) ensuring zero-downtime rolling updates for containerized Kubernetes workloads.
- Implemented Infrastructure as Code (IaC) via Terraform, drastically reducing provisioning time for cloud environments and lowering operational overhead.

Amit Learning **Jul 2022 – Feb 2023**
AI & Deep Learning Trainee *Cairo, Egypt*
- Developed and optimized PyTorch and TensorFlow vision and NLP models, applying advanced regularization, gradient clipping, and dynamic learning rate scheduling (ReduceLROnPlateau).

EDUCATION

Benha University – Faculty of Computers & Artificial Intelligence **Mar 2026 – Present**
M.Sc. in Artificial Intelligence *Cairo, Egypt*

Arab Open University **2020 – 2024**
B.Sc. in Computer Science & Artificial Intelligence *Cairo, Egypt*

CERTIFICATIONS & ACHIEVEMENTS

- **AWS Certifications:** Solutions Architect – Associate (Jul 2026), Machine Learning Engineer – Associate (Jan 2026), AI Practitioner (Sep 2025), Cloud Practitioner (Jun 2025).
- **Microsoft Certification:** Azure AI Apps and Agents Developer – Associate (Sep 2026).
- **Continuous Learning:** Deep Learning, Generative AI & NLP (39 courses, DeepLearning.AI); AI Agents & DevOps (13 courses, IBM); MLOps & Kubernetes (KodeKloud, Udemy).
- **Achievements:** ICPC ECPC Finalist – Ranked 4th at Arab Open University, International Collegiate Programming Contest (ECPC Qualifications 2024).

RESEARCH & KEY ENGINEERING PROJECTS

Reduction Ladder: Diagnosing & Resolving Code SLM Shortcut Mimicry *Ongoing R&D | Orange Innovation Labs*

- **Diagnostic Research Formulation:** Joint Orange Innovation & Benha Univ. research isolating template memorization from genuine algorithmic reasoning; established the 6-tier *Reduction Ladder* diagnostic (664 tasks, HumanEval & EvoEval) evaluating collapse bounds under semantics-preserving AST mutations.
- **Novel Mitigation Algorithm (Inv-GRPO):** Formulated and derived *Invariance-Regularized Policy Optimization*, pairing cross-perturbation rollouts with semantic consistency rewards ($\mathcal{R}_{cons}$) to break shortcut mimicry and shift collapse ($\Delta\ell^* \geq 2$) with 0ms inference latency penalty.
- **Empirical Suite & Profiling:** Benchmarking multi-arm mitigations (Contrastive SFT, AST-RL, Step-RLVR) on Qwen2.5-Coder (1.5B/7B) under edge VRAM budgets (≤8GB); tracking token efficiency, overthinking tax, and error taxonomies.

NT-sMoCo: Dense Retrieval & Contrastive KBQA | **NAACL'22 Paper** *Paper Reproduction | M.Sc. Research*

- **Paper Implementation & Extension:** Re-implemented and extended the NAACL 2022 paper (*Liu et al.*), formulating a Normalized-Temperature Scaled Momentum Contrastive framework to address critical entity sparsity in incomplete Knowledge Bases (simulated 30–100% Freebase coverage).
- Engineered a Dual-BERT dense retrieval backbone with hard-negative mining, integrated with a 2-layer Graph Attention Network (GATv2) for multi-hop subgraph entity traversal.
- Benchmarked on WebQSP, successfully replicating published state-of-the-art results (+7.9% Hits@1 over baseline) with rigorous statistical significance validation ($p < 0.0001$).

OpenQ: Autonomous Enterprise Agentic Intelligence Platform

- Developed an autonomous AI platform unifying 8 data modalities behind a natural language interface, empowering non-technical teams to query complex databases instantly and eliminating engineering bottlenecks.
- Architected a hybrid retrieval backend (Neo4j FastRP for structural traversal + Qdrant for semantic search), orchestrating workloads across 8 fault-isolated Celery workers for robust data processing.
- Implemented a self-correcting LangGraph state machine with 3-retry SQL auto-correction and persistent HITL checkpoints (AsyncRedisSaver), deployed on AWS EKS with Aurora Serverless v2.

Arabic Mental Health LLM – Distributed Alignment & High-Throughput Serving

- Engineered a zero-touch Kubeflow (KFP v2) DAG orchestrating headless Jupyter execution via Papermill, accelerating iteration cycles with full MLflow lineage from dataset validation to deployment.
- Fine-tuned Qwen2.5-3B via Unsloth (NF4 4-bit). Benchmarked 6 parallel alignment algorithms (DPO, IPO, KTO, ORPO, SimPO, RLOO), auto-merging the winning SimPO model in BFloat16.
- Configured a high-throughput serving layer using FastAPI and gRPC proxying to NVIDIA Triton + vLLM (PagedAttention, Continuous Batching), maximizing GPU utilization and ensuring low-latency API responses.

Multilingual RAG Architecture with Multi-Agent Reflection

- Designed a 5-agent LangGraph topology (Query Rewriter, ReAct Brain, Reflection Node) utilizing LLaMA-3.3-70B to iteratively detect and mitigate LLM hallucinations, ensuring highly reliable outputs.
- Optimized retrieval latency and accuracy using a 5-stage FAISS HNSW pipeline (BM25 hybrid search, cross-encoder reranking, and Redis semantic caching) to reduce compute overhead and improve relevance.

LoRA-Finetuned E-Commerce Semantic Search

- Fine-tuned dual-encoder bi-encoder embeddings via LoRA (∼90% parameter reduction) with FAISS vector indexing, achieving sub-second dense semantic product retrieval.
- Engineered a stateful LangGraph conversational workflow orchestrating FLAN-T5 for context-aware recommendations and personalized multi-turn search sessions.

TECHNICAL SKILLS

- **LLMs, Post-Training & Alignment:** SFT, RLHF, RLVR, GRPO, DPO, SimPO, KTO, PPO, LoRA, QLoRA, PEFT, Unsloth, TRL, Model Distillation, Quantization (AWQ, GPTQ, BitsAndBytes NF4), Tokenization, Chain-of-Thought (CoT).
- **Agentic AI & Systems:** LangGraph, LangChain, Multi-Agent Orchestration, ReAct Reasoning, Tool/Function Calling, Reflection, Self-Correction, HITL (Human-in-the-Loop), Semantic Caching.
- **Retrieval & RAG:** Hybrid Search (Dense + Sparse BM25), GraphRAG (Neo4j), Vector DBs (Qdrant, FAISS, Pinecone, Milvus), Cross-Encoder Reranking, ColPali, Document Chunking.
- **Inference, Serving & Optimization:** vLLM, PagedAttention, Continuous Batching, NVIDIA Triton Inference Server, TensorRT-LLM, ONNX Runtime, gRPC, FastAPI, KV Cache Optimization, FlashAttention.
- **Distributed MLOps & Cloud:** Kubeflow (KFP v2), MLflow, Distributed Training (DDP, FSDP, DeepSpeed), Docker, Kubernetes (K8s), Terraform (IaC), AWS (EKS, SageMaker, EC2, S3, RDS, Lambda), Azure (Azure AI Foundry, AI-103), Prometheus, Grafana.
- **Backend & Data Engineering:** Python (Asyncio), C++, Celery, Redis, Pydantic, RESTful APIs, WebSockets, PostgreSQL, Neo4j, SQL/NoSQL, Clean Architecture.
- **Core Frameworks & Libraries:** PyTorch, TensorFlow, Hugging Face (Transformers, Datasets, Accelerate), scikit-learn, OpenCV, YOLOv8, NumPy, Pandas, Linux/Bash, Git.